Contents

1 Introduction **1**

1.1 Music and Music Recommendation Systems 1
1.2 Mobile Applications and Cross-Platform Development 1
1.3 Project Mission and Objectives 2
1.4 Declaration of Generative AI and AI-Assisted Technologies in the Writing Process 2
1.5 How This Thesis Is Organized 2

2 Theoretical Foundations of Music Recommendation Systems **4**

2.1 Introduction to Music Recommendation Systems 4
2.2 Recommendation System Techniques 5
2.2.1 Collaborative Filtering 5
2.2.2 Content-Based Filtering 6
2.2.3 Hybrid Recommendation Systems 7
2.2.4 Context-Aware Recommendation Strategy in AfterVibes . . . 8

3 Context-Aware Recommendation Systems **10**

3.1 What Is Context Awareness? 10
3.2 User Context Dimensions in AfterVibes 10
3.3 Spotify Playback Integration 10
3.4 Benefits of Context-Aware Systems 10

4 AfterVibes Application **11**

4.1 Application Overview 11
4.2 Requirements Analysis 11
4.2.1 Functional Requirements 11
4.2.2 Non-Functional Requirements 11
4.3 Diagrams 11
4.4 System Architecture 11
4.4.1 Backend – FastAPI, Firebase, Spotify Integration 11
4.4.2 Integration Layer – APIs and Services 11

4.4.3 Frontend – React Native Interface 11
4.5 Implementation Details 11
4.6 User Guide 11
4.7 Testing and Validation 11

5 Conclusions and Future Work **12**
5.1 Summary of Contributions 12
5.2 Limitations of the Current Implementation 12
5.3 Future Improvements 12
5.4 Final Thoughts 12

Bibliography **13**

Chapter 1

Introduction

1.1 Music and Music Recommendation Systems

Music is one of the most popular forms of entertainment and artistic expression. It is a profoundly personal experience that allows listeners to connect with songs, lyrics, and rhythms on an emotional level. As a universal language, music has the power to unite people from diverse cultures and backgrounds, influencing mood, memory, and even behavior in daily life [4].

In recent years, a wide range of mobile applications have emerged, offering users convenient access to streaming music anytime, anywhere. These platforms have transformed the way people consume music, making personalized listening a seamless part of everyday life. However, despite music's emotional depth and social significance, most popular music platforms still rely primarily on static historical data or pre-generated playlists to provide recommendations. This approach often overlooks the listener's current external context, such as their activity, mood, or environment, resulting in suggestions that may feel disconnected from their present state.

1.2 Mobile Applications and Cross-Platform Development

The rise of smartphones has led to an explosion of mobile applications that address nearly every aspect of daily life, including health, education, finance, and entertainment. Mobile apps offer the advantages of portability, real-time interactivity, and constant accessibility, making them one of the most effective tools for user engagement.

The development of mobile applications using cross-platform frameworks allows deployment on multiple operating systems using a single codebase. Cross-

platform development significantly reduces both development time and cost, while ensuring a consistent user experience across devices. Moreover, modern frameworks support access to native device features and allow seamless integration with third-party services.

1.3 Project Mission and Objectives

The AfterVibes application was designed to redefine how users engage with music by providing real-time, context-aware song recommendations. Unlike traditional platforms that rely solely on listening history or curated playlists, AfterVibes responds to the user's current activity, environment, and mood to enhance their music experience.

The main objective of the project is to deliver a personalized and immersive mobile application that adapts dynamically to the user's lifestyle. This is achieved by integrating multiple data sources and offering a smooth and intuitive user experience.

Specifically, the project aims to bring several original contributions to the field of mobile music applications and recommendation systems by fusing real-time contextual data — including user activity, mood, weather, and time of day — to dynamically adjust music recommendations.

1.4 Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author, used ChatGPT by OpenAI as a writing assistant to improve spelling, grammar, and the clarity of written content. All content has been thoroughly reviewed and edited by the author, who takes full responsibility for the accuracy, originality, and integrity of this thesis.

1.5 How This Thesis Is Organized

This thesis is structured into five chapters, each addressing a critical aspect of the AfterVibes application and the technologies behind it. The structure ensures a logical and comprehensive coverage of theoretical concepts, system architecture, implementation details, and conclusions.

- **Chapter 1: Introduction** – Establishes the foundation of the thesis by presenting the context and motivation behind the project, defining its objectives, out-

lining its original contributions.

- **Chapter 2: Theoretical Foundations of Music Recommendation Systems** – Discusses various types of recommendation systems, how they work, their current challenges and limitations.
- **Chapter 3: Context-Aware Technologies in Mobile Applications** – Analyzes the key aspects of context-aware systems, including real-time activity tracking using mobile sensors, weather data integration, mood check-ins, and how these data points contribute to dynamic music recommendations.
- **Chapter 4: The AfterVibes Mobile Application** – Provides an in-depth analysis of the application, including system architecture, backend development with FastAPI and Firebase, frontend implementation in React Native, integration with external APIs such as Spotify and OpenWeatherMap, and the design of playback features. It also includes a user manual with instructions and examples.
- **Chapter 5: Conclusions and Future Work** – Summarizes the main findings and contributions of the thesis, discusses the limitations of the current implementation, and outlines potential improvements and future research directions.

Chapter 2

Theoretical Foundations of Music Recommendation Systems

2.1 Introduction to Music Recommendation Systems

Recommendation systems have become a cornerstone of modern digital experiences, enabling platforms to personalize content based on user preferences and behavior. From e-commerce to entertainment, these systems help users discover relevant items in a vast and growing sea of content.

In the music industry, recommendation systems are especially important due to the highly subjective and personal nature of musical taste. With millions of songs available across streaming platforms, delivering curated, contextually relevant music has become essential for both user satisfaction and platform engagement. Services like Spotify, Apple Music, and YouTube Music rely on recommendation engines to generate daily mixes, discover weekly playlists, and personalized radio stations tailored to each listener.

The core objective of a music recommendation system is to match listeners with content they are likely to enjoy, based on implicit or explicit signals such as listening history, song likes, skipped tracks, and user-created playlists. These systems not only enhance user retention but also increase the discoverability of lesser-known artists and songs. According to Chen and Chen [2], combining content-based, collaborative, and statistical methods can significantly improve recommendation accuracy by capturing both musical features and user behavior patterns.

However, traditional recommendation systems often focus on **what** users liked in the past, without considering **when**, **where**, or **how** they are listening. This static approach, while useful, lacks real-time adaptability and contextual understanding — two key aspects that modern users increasingly expect. This gap has given rise to a new class of systems known as **context-aware recommendation systems**, which

aim to tailor suggestions based on a user's current situation and environment.

The AfterVibes application builds on this concept, aiming to deliver music that aligns with the user's present state. By combining multiple context signals — such as physical activity, time of day, mood, and weather — AfterVibes introduces a more dynamic and adaptive approach to music selection.

Table 2.1: Comparison of Personalization and Context-Awareness Across Music Platforms [2]

Platform	**Personalization Features**	**Context Awareness**
Spotify	Daily Mix, Discover Weekly, Wrapped, On Repeat	Limited (audio mood only)
Apple Music	Listen Now, Favorites Mix, Personalized Radio	None
YouTube Music	Song Radio, Mixed for You, AI-generated playlists	None
AfterVibes (Proposed)	Real-time playlist adaptation based on activity, mood, weather, and time	**Yes**

2.2 Recommendation System Techniques

Recommendation systems utilize a variety of algorithms to help users discover content they are likely to enjoy. These systems are generally categorized into three primary types: collaborative filtering, content-based filtering, and hybrid approaches. Each category uses a different strategy to match user preferences with available content and comes with its own strengths and limitations.

2.2.1 Collaborative Filtering

Collaborative filtering (CF) is one of the most widely adopted recommendation strategies. It works by analyzing the behavior and preferences of multiple users to identify patterns. The central idea is that users who have shown similar interests in the past are likely to enjoy similar items in the future [3]. Collaborative filtering can be broken down into two major variants:

- **User-based Collaborative Filtering:** This method compares users to identify those with similar listening patterns and preferences. Once similar users are found, the system recommends songs enjoyed by one user to others in the group. For instance, if User A and User B both enjoy Songs X and Y, and User

A also listens to Song Z, then Song Z could be recommended to User B based on their shared history.

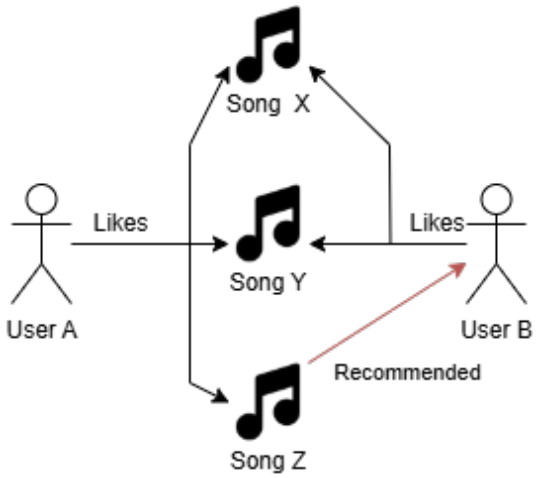


Figure 2.1: User-based Collaborative Filtering Example: Song Z is recommended to User B based on shared listening patterns with User A

- **Item-based Collaborative Filtering:** This technique emphasizes the relationships between items. The system recommends songs that are frequently associated with others the user already likes. For example, if a user frequently listens to Song X, and Song Y is commonly favored by others who also listen to Song X, the system may suggest Song Y as a relevant choice.

Despite its effectiveness, CF faces challenges such as the **cold start problem**, where new users or items lack sufficient data to generate accurate recommendations. Additionally, CF can struggle with data sparsity and scalability issues.

2.2.2 Content-Based Filtering

Content-Based Filtering (CBF) recommends items by analyzing the intrinsic characteristics of content and matching them to user preferences. In the context of music recommendation, this approach involves suggesting songs with attributes similar to those a user has previously enjoyed. Key aspects of CBF include:

- **Feature Extraction:** Identifying and quantifying attributes such as genre, tempo, mood, instrumentation, and lyrical themes. This process creates a detailed profile for each song, facilitating accurate matching with user preferences [5].
- **User Profiling:** Developing a user profile based on their listening history and explicit feedback, such as likes or ratings. This profile is then used to recommend new content that aligns with the user's established tastes [5].

- **Similarity Measurement:** Calculating the similarity between the user's profile and available songs using metrics like cosine similarity or Euclidean distance. Songs with higher similarity scores are prioritized in recommendations [5].

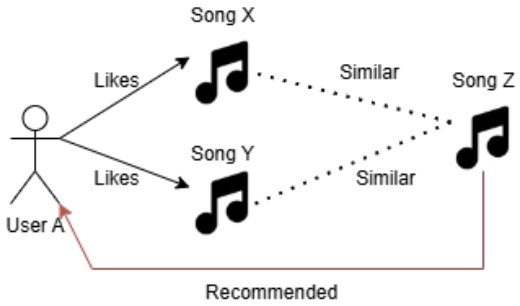


Figure 2.2: Content-Based Filtering Process: Matching song features to user preferences

While CBF effectively introduces users to new or niche tracks that align with their preferences, it has limitations:

- **Over-Specialization:** The system may repeatedly recommend songs with similar attributes, leading to a lack of diversity in the user's listening experience [5].
- **Metadata Dependency:** Accurate recommendations rely on comprehensive and precise metadata. Incomplete or inconsistent metadata can hinder the system's performance [5].
- **Limited Novelty:** By focusing on established preferences, CBF may struggle to introduce users to diverse or unexpected music styles, potentially limiting the discovery of new genres or artists [5].

Despite these challenges, content-based filtering remains a valuable technique in music recommendation systems, especially when integrated with other methods to enhance recommendation diversity and accuracy.

2.2.3 Hybrid Recommendation Systems

Hybrid recommendation systems integrate multiple recommendation techniques to capitalize on their strengths and offset individual limitations. By combining methods such as collaborative filtering and content-based filtering, these systems aim to enhance recommendation accuracy and address challenges like the cold start problem. Several hybridization strategies include:

- **Weighted Hybridization:** This approach assigns weights to the outputs of different recommendation models and combines them to generate a final list. The weights can be adjusted to reflect the relative performance or confidence in each model [1].
- **Switching Hybridization:** In this method, the system selects one technique based on specific conditions, such as data sparsity or the availability of user preferences. It ensures that the most suitable method is applied in each context [1].
- **Mixed Hybridization:** This strategy merges results from multiple systems and displays them side by side. It increases diversity in recommendations and allows the user to see various perspectives [1].
- **Cascade Hybridization:** This technique prioritizes one model to generate an initial set of recommendations and then refines or re-ranks the list using another algorithm [1].
- **Feature Augmentation:** Here, the output of one system is used as additional input features for another. For example, collaborative filtering scores may be used to enrich content-based profiles [1].
- **Meta-Level Hybridization:** This method uses the learned model of one technique as the input for another. For instance, a user profile generated by one model can be passed into another model for deeper analysis [1].

By employing these hybridization strategies, modern systems can deliver more accurate, personalized, and diverse recommendations — resulting in higher user satisfaction and retention.

2.2.4 Context-Aware Recommendation Strategy in AfterVibes

The AfterVibes application builds upon the core techniques of recommendation systems discussed previously — namely collaborative filtering, content-based filtering, and hybrid methods — by incorporating dynamic, real-world context into the recommendation process. Instead of relying solely on historical preferences or item similarity, AfterVibes generates personalized music suggestions based on multiple real-time factors:

- **Time of Day** – Adjusts recommendations to align with user energy rhythms (e.g., energetic tracks in the morning, calming ones at night).

- **Mood Input** – The user can check in with their current mood, allowing the system to adapt audio features such as valence and tempo accordingly.
- **Physical Activity** – Smartphone sensors detect activity levels (e.g., walking, sitting), influencing tempo and energy recommendations.
- **Weather Conditions** – Using weather-specific data (e.g., lo-fi on rainy days, upbeat on sunny ones).
- **Spotify Audio Features** – Tracks are filtered and ranked using Spotify's metadata (e.g., danceability, energy, acousticness) to match the real-time user context.

AfterVibes is primarily content-based at its core — analyzing song features and matching them to a dynamic user profile.

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